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Analysis of demand forecasting of agriculture using machine learning algorithm

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Abstract
The state of India was situated on fertile land and river deltas with appropriate agricultural land. In 2019, agricultural fields, primarily cropland, occupy more than 40% of the country's total surface area. Moreover, agricultural industry revenues less than 3% of the country's Gross Provincial Product (GPP). While manufacturing has become the country's major financial activity, accounting countries represent half of GPP's revenues. The objective of the study is to find a way to improve the financial profitability and efficiency of farming supply chain networks as follows: (1) Fixation of national-level targets for zonal-level groups information after assessment and forecasts that affect production and distribution of agriculture. (2) Producers risk can be reduced by directing people to multiple factory and industrialization options based on market assessment. (3) Insurance costs and reduction of bank borrowing by standardizing the connection between bankers and producers using the structure to centralise land information. (4), Stabilize the agricultural sector by looking at nearby potential destinations for production and regulation of the Public Distribution System (PDS) flow for the security stock. In this paper, the novel ML target prediction algorithm to inform the farmers about the market target product and improve the relationships between the farmer and bankers for centralizing the information about recent government plans. The crop prediction ML algorithm proposed to improve the revenue of agriculture field.

Keywords Crop rotation, ML algorithm, Farming sector · Crop cultivation · Tracking and Forecasting

1 Introduction

The Government of India has laid an objective of doubling farmers' income by the year 2022 through several schemes like the National Mission for Sustainable Agriculture and the National Mission on Agricultural Extension and Technology (Abraham & Pingali, 2021; Ceballos et al., 2019; Shah & Wu, 2019). The modern farmer who cultivates unconventional

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farm produce lacks in calculating the demand and supply analysis which fails in predicting the season demand where the commodity faces price fluctuation and leads to lower-income (Dimkpa et al., 2020).

Today's farmers use the technology because the benefit of time and money is well known. In agriculture which is based on the uncertainty of climate, crop failures on pest attacks, and is highly risk-oriented the coverage of insurance will be automated according to the area of the crop covered and the farmer individuals as recorded (Lowenberg-DeBoer et al., 2020). This data entry process made on the ground level should be recorded based on the availability of the groundwater level, the water requirement of the crop, the rainfall pattern, etc.; of the district (He et al., 2018; Rajput et al., 2021). The data will be collected as a bottom-up approach with several factors determining the annual target and the land details from the satellites after matching (Iyer et al., 2021). The analysis will be multidimensional interlinking of major Government departments and NGOs to improve the growth of the agriculture sector (Jha et al., 2021).

Recommendation System (RS) is the general category for Internet applications that predict client responses based on a variety of alternatives. It can be seen in a wide range of more current applications which provide users with access to vast databases. Algorithms usually present a client simply a listing on proposed products and an estimate on whether highly someone would like everything (Tröster & Küblböck, 2020). Machine Learning (ML) algorithms help customers select the right things and facilitate the localization of the desired things within a collection. Several varieties of prescriber algorithms have been used on the web in different areas, with each version attempting to correctly predict the consumer's desire. Furthermore, several researchers have studied multiple aspects of RS such as adaptability, etc. (Mehrotra & Sergeyev, 2021).

A volume that data and things have become massive, resulting in data overload. Finding what a customer was looking for has become an enormous challenge (Nedumaran & Ravi, 2019). The search technologies helped to resolve the parts problem, but no customization of the data was provided. As a result, programmers turned to a RS for help (Sabu & Kumar, 2020; Salem et al., 2018). The RS software enables the selection and classification of objects and data. Researchers implement their perspective on the population, as people can help members of this group identify relevant documents with the greatest success from a potentially excessive set of options (Lukyanova et al., 2020). It uses a variety of techniques that have been split into 3 categories: information algorithms, collaborative filtration devices, and mixed RS (Sardar et al., 2021).

To overcome the lacuna in agriculture from the farmer's point and to suggest the roadmap along the lines of developed countries where cultivation takes place in a larger area compared to scattered land spread area cultivation in our country (Itelima et al., 2018; Singh & Burman, 2019). To overcome the unplanned cultivation of crops and instability of demand forecast leads to price fluctuation of agriculture commodities (He et al., 2020). With the means of Geographical information system for collecting the data and connection with farmers, the prediction of agriculture cultivation forecast is possible at the fingertips (Tigabu & Gebeyehu, 2018). The growth of agriculture is the strong foundation for the development of the nation's economy on employment generation, industrial growth, etc.

2 Materials and methods

Data of the six major factors that determine the target fixation has to fetch into application from various sources. All these have to be summed up from zonal to the national level. These factors include:

(i) *Land Relief (L)* data give a clear-cut view of what crop is suitable for the given land. Details like whether it's a plain or valley are later sources available nearby or not etc.
(ii) *Water availability data (W_A)*
(iii) *Temperature range (T)*
(iv) *Soil characteristics (S)*
(v) Data about the *export and import of agricultural products (X)*
(vi) *GPS tracking using satellite images data*

All these data will be collected in a bottom-up approach from districts, then state and clustered at the national level from farmers, various government departments mentioned in the collaboration, and traditional data for training purposes.

All these will be sent in through an API from a database to the app structure. We will set this API either using the Lead Squared platform or using a Django REST framework depending on the internal structure. The data collected from various departments and sources has been clearly illustrated in Fig. 1.

3 Proposed methodology

The algorithm used in the framework produces two predictions i) Now the data used in a multiple linear and a logarithmic regression of the supervised learning for analysis of the given data to predict the target and recommend the suitable crop using the respective field data. Then together the clustered formed through virtual network. Then the specific field data from analysis using various graphs and segmentation using components like Keras, pandas, NumPy, seaborn, sci-kit learn, and matplotlib. Now the data used for prediction of the trend i.e.; risk analysis of farmer by dictvectorizer using one-hot label encoding. Also, the data from satellite images undergone aggregation using noise amplification and then it matched the process using a Convolution Neural Network (CNN). The flow of the algorithm process has been illustrated in Fig. 2.

Let's analyze, *L*—Land Relief factor, *R* is Rainfall and *G* is groundwater content

$$W_A = R + G \tag{1}$$

T—Temperature (Climate).

S—Soil Characters (texture,colour,alkalinity/acidity,depth etc.).

Since most of these values are numeric directly, hence we label them using one-hot encoding and assign the values to use in analysis and prediction. Now, these symbols become L_f, R_f, T_f, and PHP, respectively.

The data from the Import and Export balance sheets and the PDS flow for predicting which we represent using X (IE_f and PDS_f), now we represent the data for the different seasons:

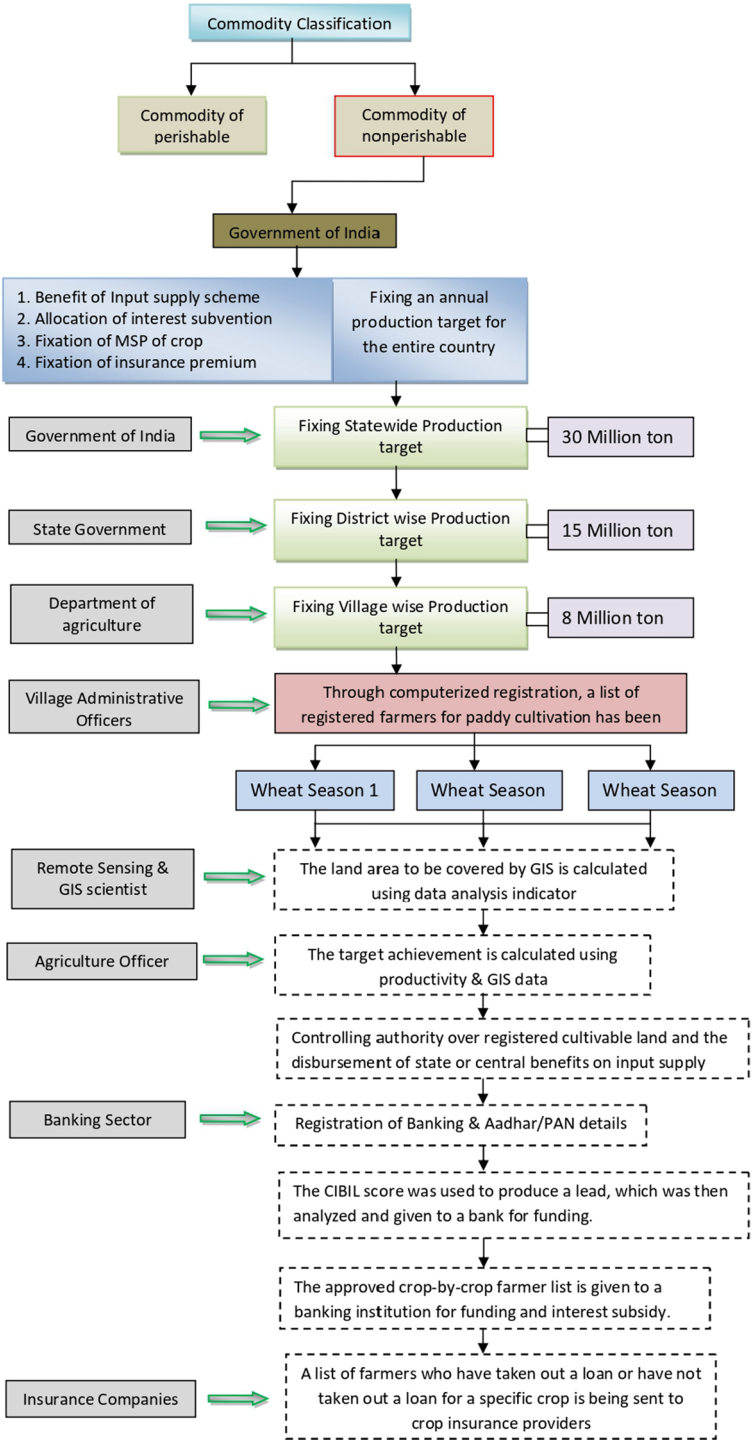


Fig. 1 Flow of collection and other processes

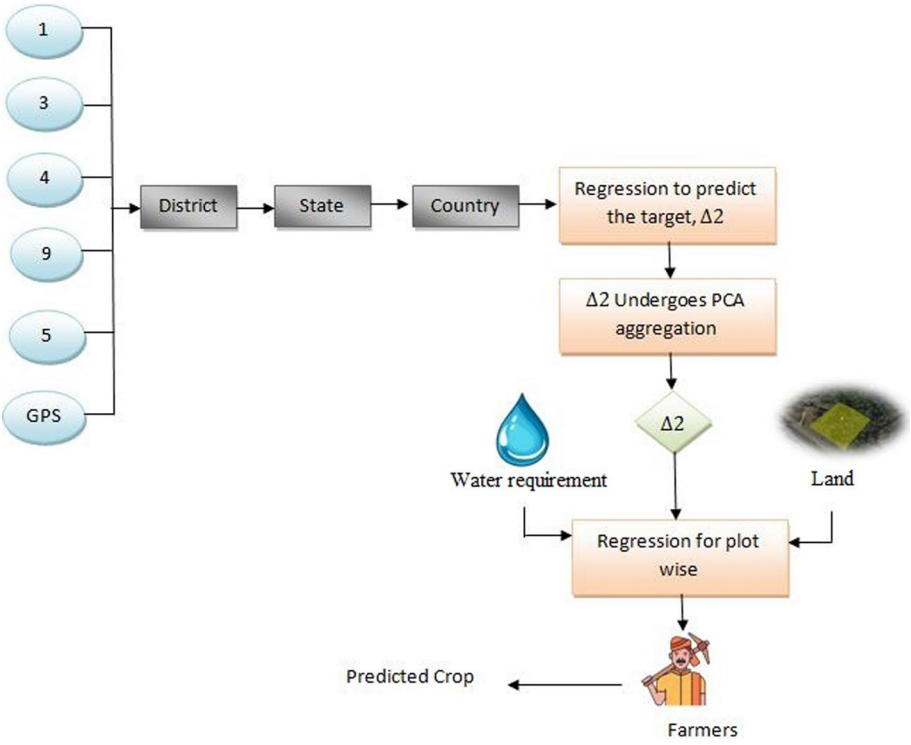


Fig. 2 Process flow diagram

3.1 National level target prediction algorithm

Input: six factors

Process: Summation of Zonal wise analysis using Linear regression

Output: National-level target

Basically, f will be a combination of the kharif and rabi season,

$$f = f_{\text{kharif}} + f_{\text{rabi}} + f_{\text{other}} \tag{2}$$

the next step is to finalize the total for the whole country which is actually a cluster of different zonal levels:

$$T_1 = f(L_f) + f(W_A) + f(T_f) + f(PH_f) + f(X) \tag{3}$$

Adding all the zonal levels we get,

$$T = T_1 + T_2 + \cdots + T_n \tag{4}$$

$$\sum T = f\left(\sum L_f\right) + f\left(\sum W_A\right) + f\left(\sum T_f\right) + f\left(\sum S_f\right) + f\left(\sum X\right) \tag{5}$$

Now we have to add the sixth feature for dataset comparison and learning using the satellite images of the previous year or season:

$$\sum T = f\left(\sum L_f\right) + f\left(\sum W_A\right) + f\left(\sum T_f\right) + f\left(\sum S_f\right) + f\left(\sum X\right) + f(\text{GPS}) \quad (6)$$

With this we can now predict the target for the current year or season or fiscal year for the budget using the regression: X_train, X_test, y_train, y_test = train_test_split (X, y, test_size = 0.3, random_state = X) Using the graphs, we analyze it through 70–30 train-test data sets from the label encoded input. The required prediction part follows next:

$$\text{Theta} = \text{model 2.coef} \quad (7)$$

3.2 Crop recommendation algorithm()

Input: National Target (T) and field-specific factors

Process: Prediction of the suitable crop to the farmer as per his field data

Output: Crop suitable to cultivate

Now, the main aim is to take it to the farmer and support him through a prediction of the crop to reduce his risk and increase productivity, hence out of the predicted set of crops, we move more specific and dynamic through a strong suggestion directly to the farmer. For that, we do a Principal Component Analysis (PCA) using aggregation and get the value for ΔT

$$\Delta T = p(T) \quad (8)$$

Now, there are two important factors that we get from farmers for land specific analysis: W_r (Water requirement and available for the crop) = variation of crop based on the availability of water to the field in various seasons throughout the harvest year and LS (Land Size the farmer wants to cultivate) = variation of crop occurs due to the size of the land on which the crop is sown again using regression,

$$P_C = f(\Delta T) + f(W_r) + f(\text{LS}) \quad (9)$$

P_C is the predicted crop based on the specific field analysis of the alternate crop cultivation probability. Again the values are reverse encoded and the values and features of the factors get displayed with their role in the prediction. Later using the Flutter UI/UX the target production and stable amount of products are displayed. The crop is to be sown in the mentioned field without any risk by the farmer that will support the water availability for the specified land size.

The following seems to be the connection between the smallholder farmer's ideal planted amount and bank financing (q* b), guarantor finance, and financing decision:

(1) q_b^* isalwayslessthan $q_{g,}^*$ thatis, $q_b^* < q_g^*$

(2) There exists critical threshold $\bar{t}$ suchthat $q_b^* < q_g^*$onlywhen $t < \bar{t}$, where

$$\bar{t} = \begin{cases} c + 1 - \sqrt{c}, & \text{if } c < \frac{1}{4} \\ 1 - c, & \text{otherwise} \end{cases} \quad (10)$$

cost $c < \frac{1}{2}, q_b^*$ isalwayslessthan q_d^*(therfore $q_b^* < q_d^*$.

(3)When the production Otherwise, there exists a critical threshold $\bar{\bar{t}}$ such that $q_b^* < q_d^*$ only when the unit commission feet $< \bar{\bar{t}}$, where

$$\bar{\bar{t}} = \begin{cases} \frac{c}{2} + 1 - \sqrt{c^2 + 4c}, & \text{if} c < \frac{1}{6} \\ 1 - 2c, & \text{otherwise} \end{cases} \tag{11}$$

When compared to the optimal production volume in a centralized capital-unconstrained distribution network QC, q* b < QC always holds q* g > QC whenever the unit commissioning charge t < c would be less than 1/2. Because the farmers do have a great incentive to create due to the comparatively higher profit when the units commissions charged it was minimal, the network could impose a higher rate of interest to collect the excess from the farmers (Mehrotra & Sergeyev, 2021). When it becomes great, however, the producer was hesitant to grow extra due to low profitability. The platforms would then cut their rate of interest, which might be much reduced than its initial economic cost, which is designed to motivate the producer to increase his or her output.

4 Results and discussions

4.1 Automation of the sector entries and predictions

Creation of a user-friendly app using flutter and a voice chatbot mechanism using python and angular 6 for analyzing the factors and predicting the target and guiding the farmers. Here the banking sector will be integrated for a standard flow and setting up easy insurance and loan setup schemes. Through the app, the entry and editing of the data will help in maintaining standardized sector records without any manipulations and data loss. These details and the prediction from our algorithm will be stored in the government database. The complete flow of data has been illustrated in Fig. 3.

CNN can be used for grouping, evaluation, and forecasting The gathered and analyzed satellite images were 2D arrays that must be handled using the Keras. Layers Convolution 2-D layer module (Sardar et al., 2021). As indicated in Fig. 4, the four functions in the process of creating CNN were multiplied, sharing, flatness, and complete connectivity. The CNN stages were performed in the following order. (1) In the training examples, the initial phase of CNN was fully connected layers activities, which have been conducted utilizing imported Conv2D from Keras. Layers. A convolutional 3-D array can be used when working with movies in which the third aspect would be a period. The convolutional algorithm includes 4 assertions: the amount of filtering, the shape of every filtering, the input structure, and also the image kind, and also the transfer functions. There seem to be 32 filters in all, including one with a 3×3 size. The input image structure in CNN has a frequency of 64×64 pixels and is RGB color. Back and white or RGB images were used. (2) The image size can be reduced more than feasible in the second phase of CNN using pooled techniques. It was an effort to cut down on the total number of instances in the levels that follow. Importing the MaxPooling2D library from Keras. Layers were required for the sharing operations stage of creating CNN.

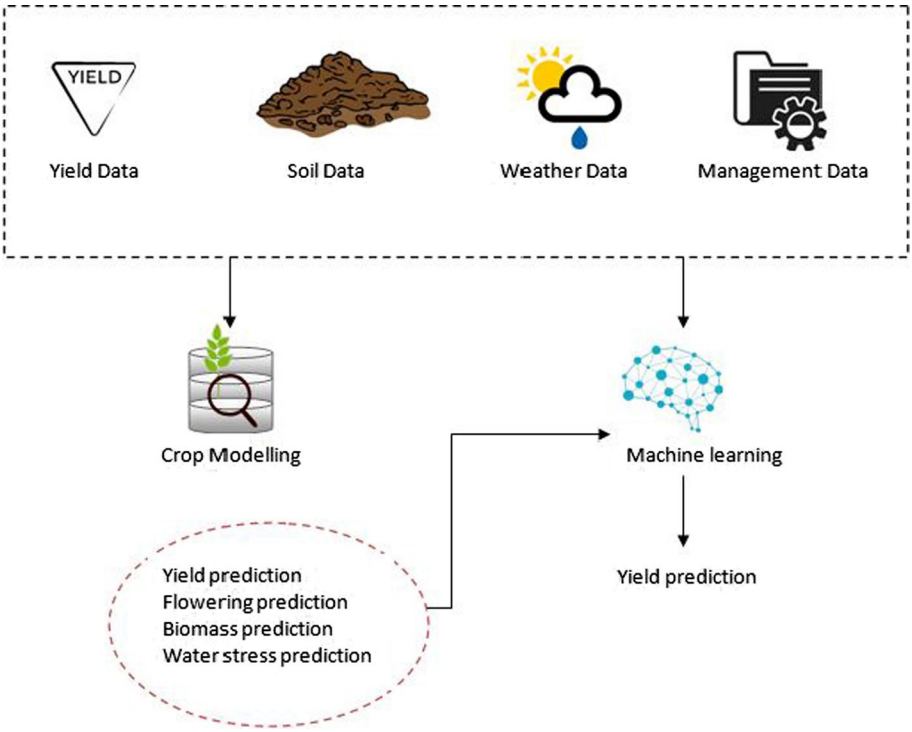


Fig. 3 Data flow and mechanism diagram

Different pooled processes, including Min Pooling, Mean Pooling, etc., were supported by the Maxpooling method. The MaxPooling process in CNN selects the greatest pixels from the appropriate area of interest (3) In the flattened operational stage, every one of the pooled images of the second phase was turned into a 1D single continuous vector. This phase of creating CNN necessitates the use of the Flatten program from Keras. Layers to turn all of the MaxPooling investigation's output 2D arrays into single sustained straight vectors. (4) The Density module from Keras. Layers were required to add a convolutional layer. The packed method takes two assertions: measures and also transfer functions. The number of nodes in each layer is indicated by the meters argument. The argument quantity of the units varies based on the overall number of incoming and outgoing connections.

4.2 Agricultural market analysis and risk assessment

Figure 5 depicts how manufacturers connect both input and output marketplaces during the marketing process, whenever there is increased availability with consumers. Producers could get credit and inputs including seeds, pesticides, fertilizers, labor, and technologies from the input markets. Within those marketplaces, both official and informal participants engage. Bankers, governmental agencies, and retailers were formal actors who can provide loans, inputs, and agricultural extensions. Moneychangers, traders, and certain other middlemen step in, to provide these services when official institutions were unavailable.

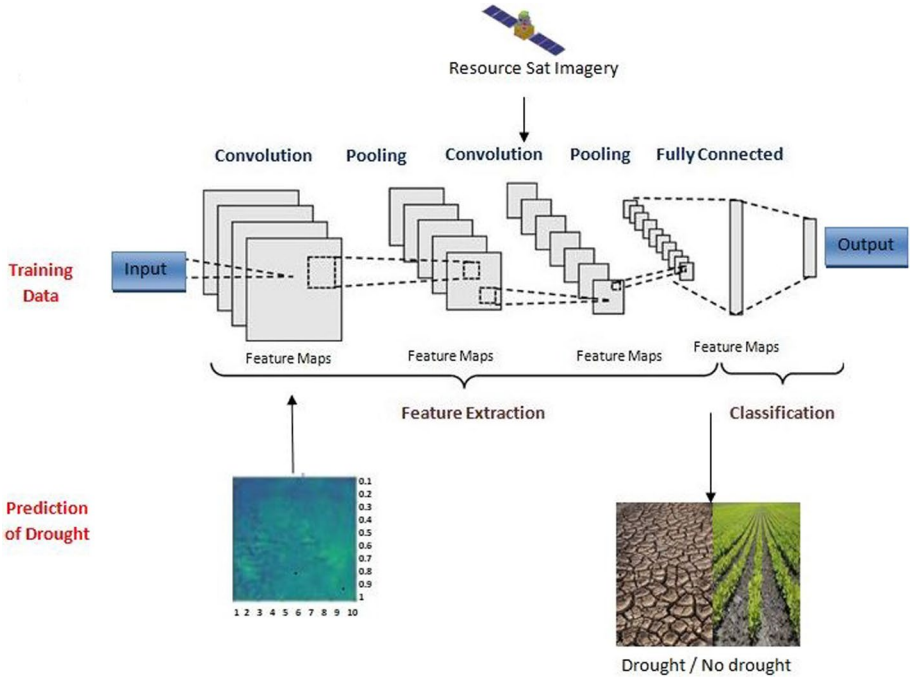


Fig. 4 The conceptual method

Producers can sell fresh pulses and buy processed pulses for family consumption at these marketplaces, which include both purchases and sales. RPMs service a small number of villages and seem to be useful for farmers having a small surplus who would be hampered by distance (Abraham & Pingali, 2021). Manufacturers might, however, obtain a better value at these marketplaces than they would if they sold to local merchants. Whenever quantities of the product were small or distant marketplaces were great, the principal customers in these marketplaces were village community traders who purchase at the farm gate. The entirety of the controlled market exchange for agriculture products was conducted by secondary markets or AMPCs.

The relationship between agricultural commercialization and access to markets creates an incentive for the availability of the agricultural product. The demand responsiveness finding shows that the regional response to pricing was mainly inelastic, but it would be impacted by competing crop output and prices, as also irrigated and monsoon availability. The agricultural motivation to invest in developing, high-quality inputs, and infrastructures like irrigation were based on price realization and market employee engagement. Costs and accessibility would continue volatile if pulses agriculture continues semi-commercial, with limited farmers' participation in markets and supply adjustments occurring only when values were excessively high. It would be critical to address market entry challenges by lowering proportionate and variable transaction fees. Researchers advocate organizational measures to better encourage aggregating methods and market reforms to address problems of transaction fees as vital to increasing commercialization of pulses farming evidence that was presented in this research.

Aggregating systems and vertically organizational alignment were two potential approaches that could assist small farms to overcome their limitations. Farmer Producer

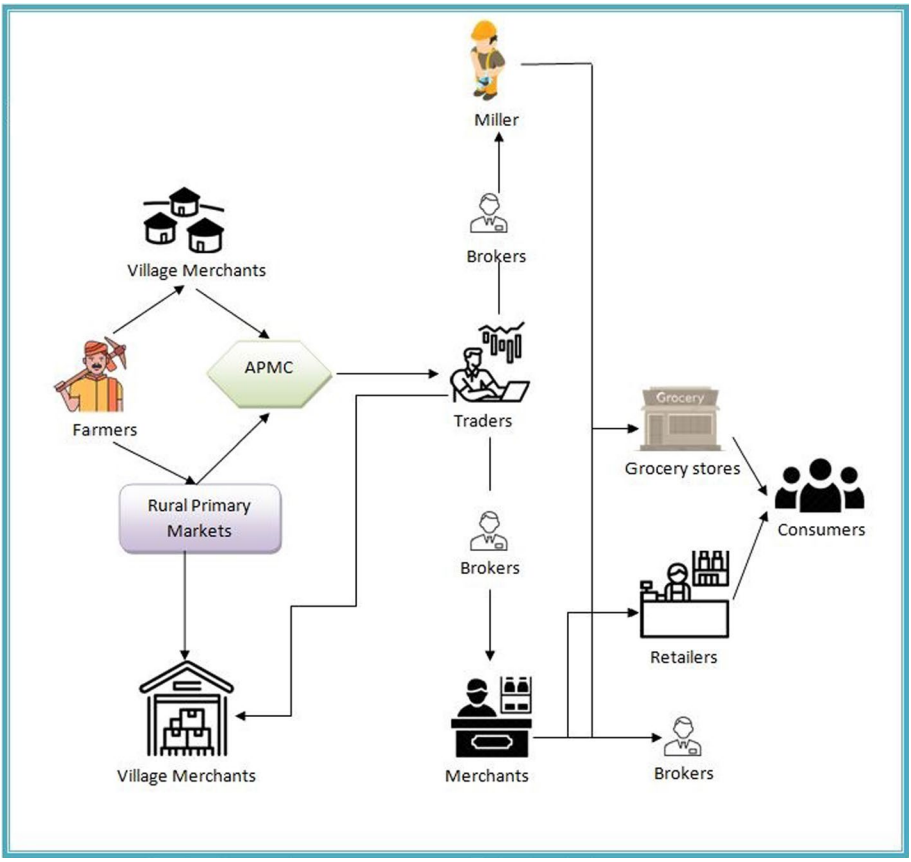


Fig. 5 Marketing networks, primary and secondary

Associations and cooperatives were instances of aggregating systems in which marginal and small farmers band together to pursue a shared goal. Small farm aggregating projects have been demonstrated to enhance access to markets and inputs like seed, loans, fertilizer, and herbicides, and also technologies and agricultural extension. Lower management policy, premium price realization, and also the ability to obtain a and sale without the need for a transfer have all been benefits of warehousing. Smallholder involved in these networks faces considerable infrastructural and scale-related issues. Producers were concerned about access and connectivity to storage and pickup places. This could be addressed by combining aggregate models of various advertising alternatives.

4.3 Agriculture sector by analyzing the potential markets

Participants were encouraged to depict the intricacy of interrelated issues by identifying the sources and consequences of the highlighted issue using the issue tree method. Users acquired a deeper grasp of the issue and also its environment through the issue tree study, with the help of this study group, which helped provide insights for offering possible approaches. In a similar manner to a concept map, but with more organization, issue

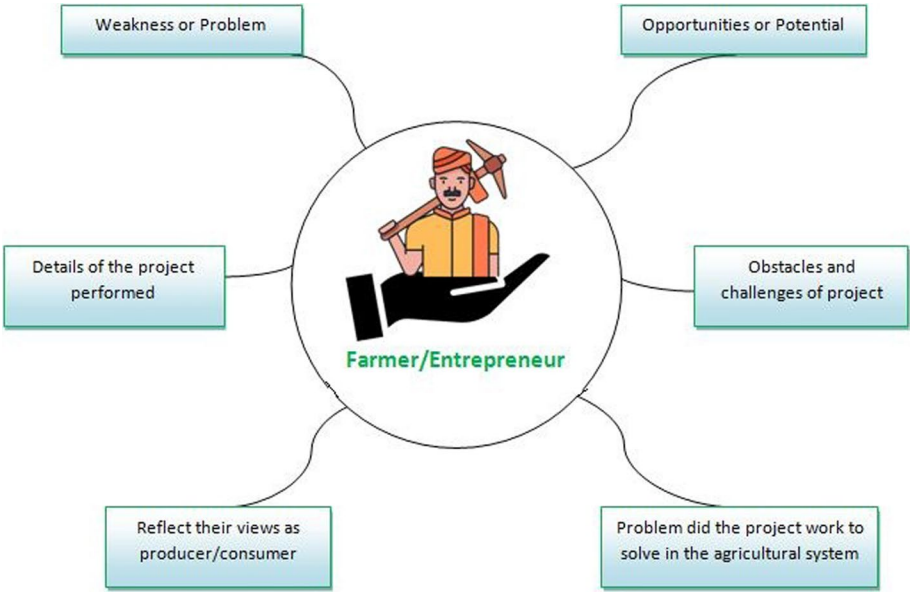


Fig. 6 Structure of the study

tree research aids in the discovery of answers by sketching out the structure of cause and effect surrounding an issue. The authors used a qualitative research methodology centered on the DPSIR framework to assess the findings. This allows us to gain a better grasp of the demands of the participants and also to look for different alternatives in the framework of India Province. Figure 6 depicts the outcome of the exercise.

Many different projects, such as co-invest agricultural platforms and a younger intelligent producers' initiative, have lately been launched to encourage contemporary agricultural technology and information exchange in terms of manufacturing and sales. Nonetheless, resulting from land instability, insufficiently incentive marketplaces, and inadequate supportive policies, the available information does not successfully pass to the number of small and medium producers at the present level that could lead to performance. As a result of these pressures, smaller people are compelled into a poverty trap, where producers receive a small amount of capital whereas the intermediary earns even more shown in Fig. 7.

4.4 Relationship between banks and farmers

Generally, the optimum actual production grows only slightly as t rises. Remember from Lemmas 1 and 2 something under banker and guarantee funding, the optimum actual production drops with t. As an outcome, as seen in Fig. 8, there seem to be units commissions, a charge threshold below/above which even the optimum actual production under financing was less/greater than it was under bank loans.

It's important to mention that farming overcapacity seems to be a common occurrence and now has long been a matter of discussion. Business loans or public subsidies were among the most commonly discussed drivers of change. The establishment of a credit line

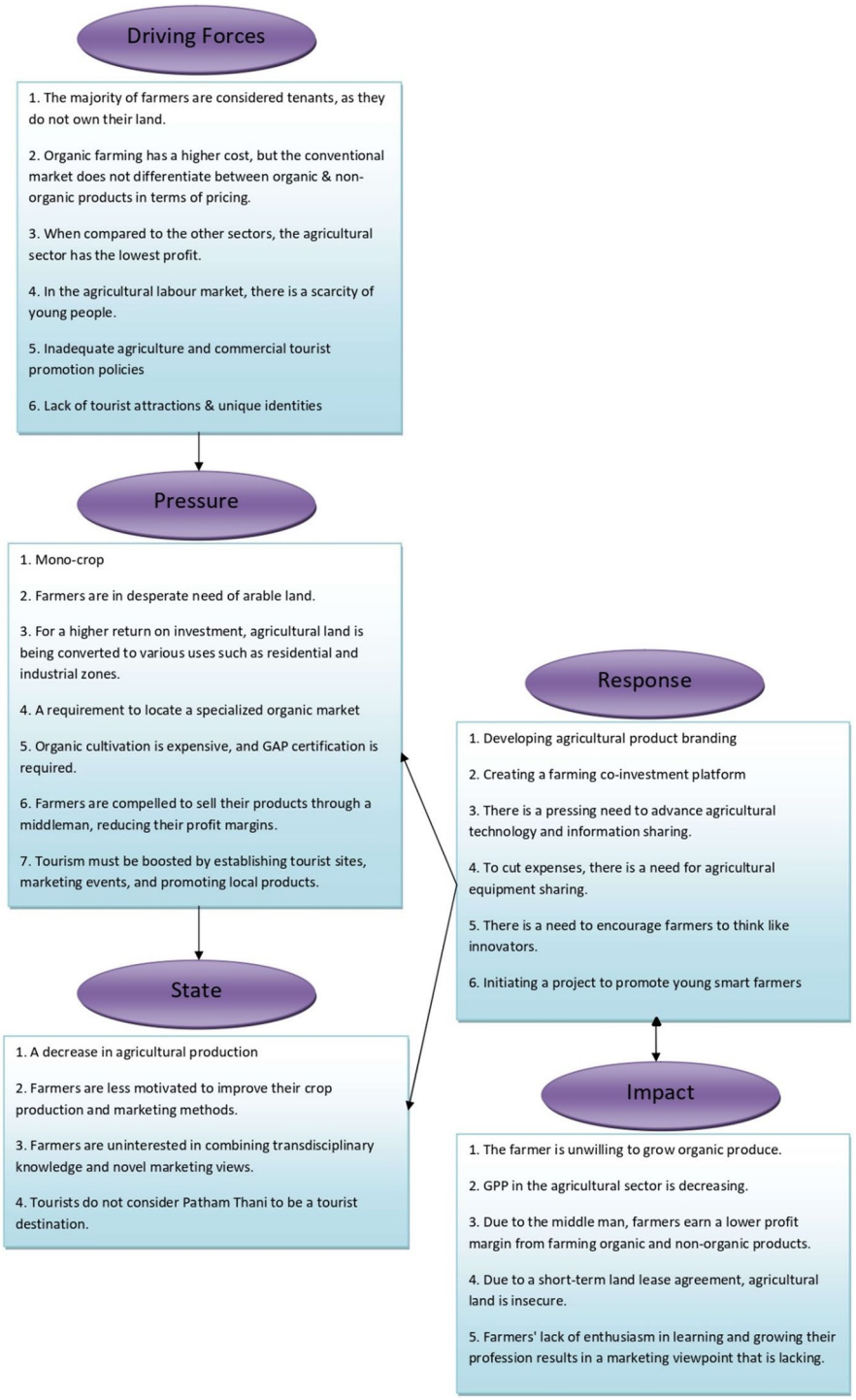


Fig. 7 Proposed framework

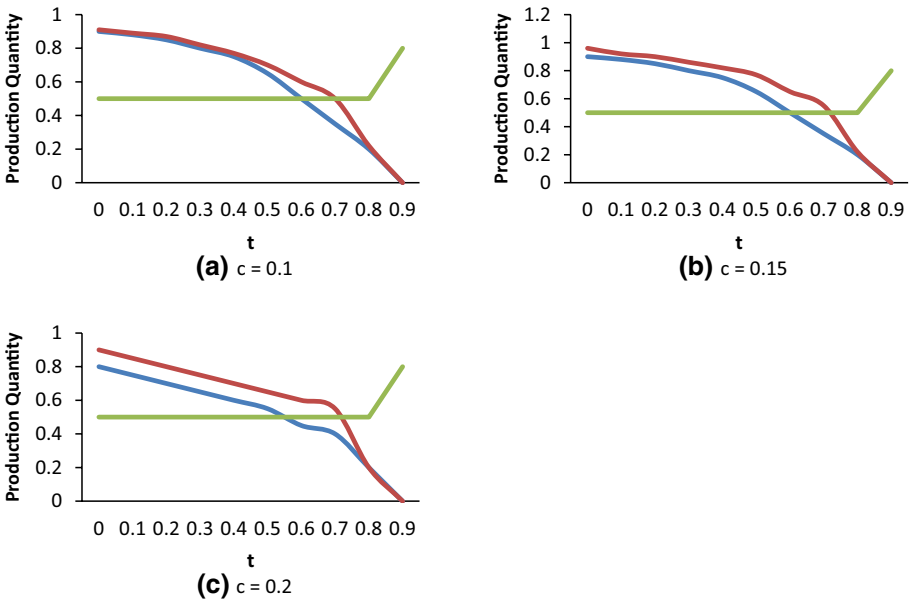


Fig. 8 Optimal variety of manufacturing product

seems to be a viable solution to this issue. That seems to be, by setting a loan amount, the intermediate platforms, which would be concerned with the overall supply chain's success, could actively manage the landowner's production volume. With this constraint, the farmer would be unable to receive as much financial assistance as he desires, potentially alleviating or even removing the issue of excess production. We analyzed the equilibria results organized into three finance structures in the monitoring and evaluation process, provided that they have been all available. The intermediate platforms, on the other hand, provide both direct and guaranteed funding, and they should be able to specify if and whether to supply every one of these types. In this chapter, humans look at a situation in which the intermediate platforms lead the charge and decide on the financial type ahead of time. The producer then chooses the finance option that maximizes his income based on available funding sources. In this context, the system's financing structure competes directly with bank funding, particularly when it comes to direct funding, where the platforms also set the rate of interest. That seems to be, by performance tuning possible solutions including the rate of interest, the platforms could affect the farmer's choice. It's also worth mentioning whenever a smallholder landowner's creditworthiness isn't good, bank funding isn't an option. the producer must then seek a loan through either straight or guarantor finance. When the farmers, on the other hand, would be credited-worthy, they could get a loan from the bank or an intermediate platform. The accompanying notion highlights the intermediate system's funding style choices while working with a credit-worthy and non-creditworthy productive, accordingly.

The median plant covering Kolar throughout the research period from 2015 to 2021 was depicted by the NDVI. Figure 9 illustrates the situation. The NDVI curve was zigzagging in form, with a progressive increase in NDVI values followed by a gradual reduction in NDVI values. During the stipulated period, it ranges between 0.2 and 0.5 for the study region. The occurrence of the drought was related to the substantial discrepancies among

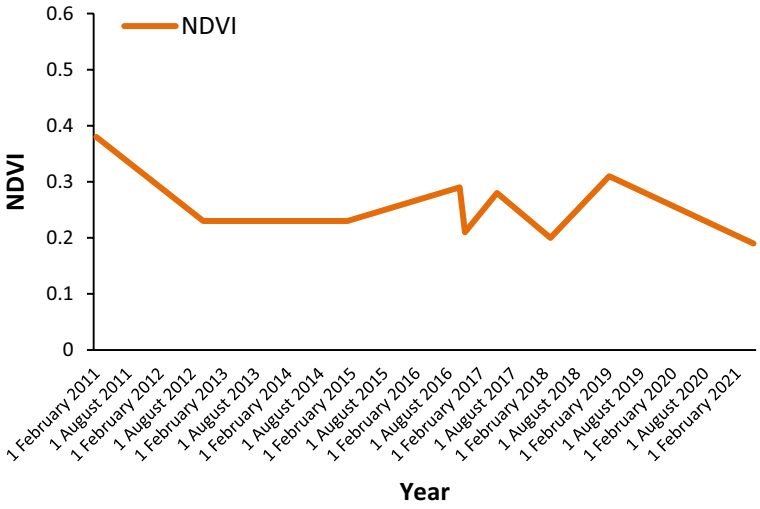


Fig. 9 NDVI results for Bands 3 and 4 in Kolar district

the NDVI median, maximum, and minimum values. The region with the lowest NDVI has no drought, whereas the areas with the highest NDVI have a persistent drought. The NDVI median correlates to the curve, which implies a near-zero frequency. With NDVI values ranging from 0.4 to 0.6, a specific region has experienced drought. In much of the area, the highest NDVI of 0.6 implies mild drought recurrence, whereas, in the remaining sections, it must be 0.4. Regarding latitudes, there seems to be a variance in the steady decreasing propensity in moderate drought occurrence.

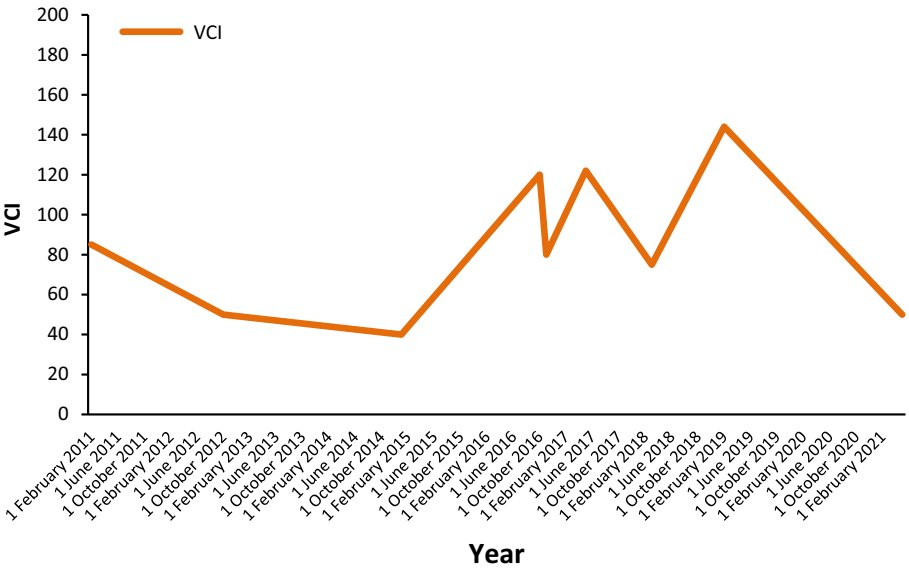


Fig. 10 VCI values for Band 3 and Band 4

The VCI determines the severity of dryness in a certain area. People activities in the region, sunshine length, weather, wind speed, and air pressure have all been elements that influence the VCI. Figure 10 shows the VCI for every frequency image. The VCI has been reported to decline at times, implying that the harshness of the droughts has risen over time. A lower VCI number implies a poor vegetative condition, whereas a greater VCI value shows a healthy vegetative condition. For the development of a CNN-based forecasting model, the VCI information of all spectral images has been used. Supervised learning and perhaps a test dataset were required for CNN. Images from bands 3 and 4 were gathered, with 80% of the images going into the training examples and 20% going into the validation data. The images from the training and validation were picked in such a way that they proposed dryness and also no drought. In the Kolar area, a total of 35 satellite images was gathered and deemed a collection.

The CNN classifier was constructed to forecast drought and also no drought that used a training data set, whereas the system testing utilizes a training dataset to forecast drought. The producing better quality by a tiny margin following executing the optimization testing. The previous versions were found to be extremely efficient, allowing the accuracy difference between the improved methods to be properly identified. The variable region in the CNN algorithm matches the time change displayed in Fig. 11. Despite the relatively modest data gathering examined, this is quite fair, with roughly 96 percent accuracy.

Due to a lack of distinguishing characteristics, the economic and tourism industry has trouble presenting tourist destinations and marketing local goods. In terms of passenger expenditure and impression of a tourist attraction, the current state of tourism activities in this region would be in a slump. To address this constraint, the respondents acknowledged that information sharing and creativity government-university collaboration could aid in the promotion of tourist destinations, the development of novel programs, and also the launch of tourism promotions.

Humans can boost the tourist and sectors of agriculture by adding an agricultural-based tourism strategy or community-based tourist, which members heated up this topic during the focus group discussions, by creating a powerful opinion of students in India about the country of farming.

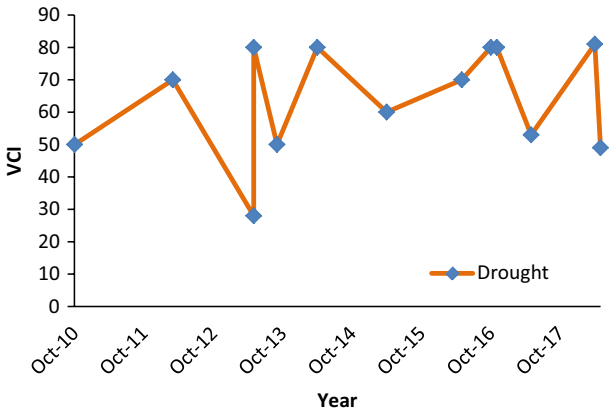


Fig. 11 Drought analysis

5 Conclusion

Drought in agriculture has severe consequences for human life and must be effectively controlled. There have been many algorithms designed to predict agricultural drought, however, performance completely depends on the method employed the information type. Many agricultural areas have been converted for other uses, including the manufacturing sector and new construction initiatives, due to the decreased margins of agriculture revenues. Simultaneously, today's local farmers face challenges such as land tenancy, inability to reach the produce market, and a lack of incentives to transition their monoculture into a novel strategy of farming. For satellites graphics drought forecasting using CNN, there can be only a few models available. The proposed study is based on satellite images and employs the Machine Learning algorithms to enhance accuracy by up to 96 percent. As a consequence, both the farmers and the entire supply chain benefits from increased social benefits. That seems to be, increasing the product's socially responsible concern could result in a win–win-win situation for all involved parties. In this paper, the novel ML target prediction algorithm to inform the farmers about the market target product and improve the relationships between the farmer and bankers for centralizing the information about recent government plans. The crop prediction ML algorithm proposed to improve the revenue of agriculture field. The proposed model useful in real-world research to determine the acceptance of the finance forms reported in this research. The new multi-cropping model, and a creative attitude as a farmland entrepreneur, must be nurtured among traditional farmers. Future agricultural policies and plans may have to incorporate the idea of "agriculture professional" to handle agricultural information and pass it on to the next generation.

Authors contributions All authors discussed the results and contributed to the final manuscript.

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Availability of data and materials Not required for this study.

Declarations

Conflict of interest Authors do not have conflict of interest.

Ethical approval Ethics approval was not required for this study.

Consent to participate Authors understand that all information I provide for this study will be treated confidentially.

Consent to publish Authors consent for the publication of identifiable details, which can include photograph(s) and history details within the text to be published in the above Journal and Article.

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